Supplementary Material
Does Synthetic Data Preserve Cluster Structure? A Factorial Evaluation of Partitioning and Hierarchical Clustering on CART-Generated Data

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Abstract

Supporting figures and tables for the main article.

1 Recovery of the number of groups by cluster count and separation

Figure S1 presents the recovery rate of the true number of groups as a function of the true cluster count k and the separation σ, for both algorithms. It carries the same information as Figure 3 of the main article in a different layout, making the $k \times \sigma$ interaction directly legible: recovery collapses in the lower-left quadrant (many groups, little separation) and is essentially perfect in the upper-right.

2 Agreement in the estimated number of groups between original and synthetic data

Table S1 cross-tabulates the number of groups estimated from the original data against the number estimated from the corresponding synthetic data, stratified by separation. Unlike the recovery rate in the main article, which scores $\hat{k}$ against the construction-time ground truth, this table compares the two estimates with each other and so isolates the effect of synthesis from the difficulty of the clustering problem itself.

The pattern confirms the expectation that at high separation the mass concentrates on the diagonal: exact agreement rises from 68.4% at $\sigma = 0.1$ to 98.2% at $\sigma = 10$ for k-means (Ward: 64.6% to 97.5%). In the overlapping regime the off-diagonal mass is not

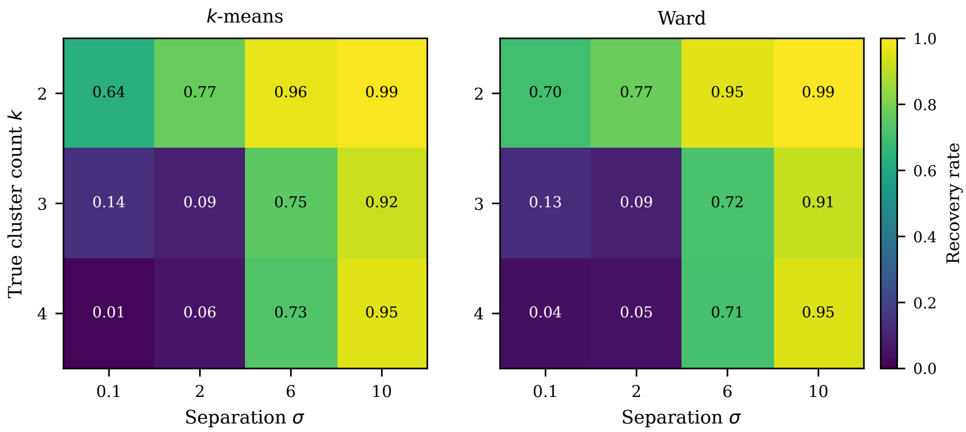

Fig. S1 Recovery rate of the true number of groups by true cluster count and separation, for k-means (left) and Ward (right). Cell values are proportions over the full design.

symmetric. At $\sigma = 0.1$ the synthetic data yield a *lower* estimate than the original in 15.2% of comparisons and a higher one in 16.4%; the largest single off-diagonal cell is the original data supporting $\hat{k} \geq 5$ while the synthetic data support $\hat{k} = 2$ (8.4%). That is, when the original data contain weak, diffuse structure that the silhouette criterion resolves into many small groups, CART synthesis tends to smooth it away into a single coarse split. The tendency disappears once the groups are genuinely separated.

Table S1 Agreement in the estimated number of groups between original and synthetic data, by separation. Rows are $\hat{k}$ from the original data, columns $\hat{k}$ from the synthetic data; cells are percentages of all comparisons at that separation (each block sums to 100%). Estimates of five or more groups are pooled for display, but the agreement column is computed on unpooled values. Both data sources are searched over a fixed range $\hat{k} \in \{2, \ldots, 8\}$, so the comparison is symmetric; under the narrower per-scenario range used for the recovery rates in the main text, agreement is higher in the overlapping regime (75.1% and 67.6% at $\sigma = 0.1$, 79.7% and 75.9% at $\sigma = 2$, for k-means and Ward respectively) and essentially unchanged at $\sigma \geq 6$, where the two conventions differ by at most 0.1 percentage points. $n = 180{,}000$ comparisons per separation level.

		$\hat{k}$ recovered from synthetic data (%)				Exact
σ	$\hat{k}$ (original)	2	3	4	≥ 5	agreement
k-means						
0.1	2	50.9	2.1	0.5	3.8	68.4%
	3	2.2	10.0	0.5	4.0	
	4	0.0	0.0	0.0	0.0	
	≥ 5	8.4	1.1	0.6	16.0	
2	2	62.9	3.4	1.6	4.2	76.4%
	3	1.2	8.6	0.7	2.2	
	4	0.3	0.4	0.9	1.2	
	≥ 5	2.5	0.9	0.9	8.0	
6	2	38.1	0.9	0.3	0.1	92.6%
	3	0.7	29.1	1.1	0.1	
	4	0.5	2.4	24.7	0.8	
	≥ 5	0.0	0.1	0.2	0.8	
10	2	34.8	0.2	0.2	0.4	98.2%
	3	0.2	31.7	0.3	0.0	
	4	0.0	0.2	31.7	0.3	
	≥ 5	0.0	0.0	0.0	0.0	
Ward						
0.1	2	58.3	5.0	2.0	6.9	64.6%
	3	3.6	4.6	1.1	3.5	
	4	1.2	0.9	0.4	0.9	
	≥ 5	6.3	1.3	0.6	3.4	
2	2	65.7	4.7	2.1	5.3	73.8%
	3	1.1	6.4	1.0	2.1	
	4	1.1	0.5	0.4	1.4	
	≥ 5	2.6	1.0	0.8	3.9	
6	2	37.1	1.0	0.2	0.1	88.2%
	3	1.9	27.3	1.6	0.3	
	4	1.8	2.7	23.6	1.3	
	≥ 5	0.3	0.5	0.1	0.3	
10	2	34.8	0.6	0.1	0.0	97.5%
	3	0.3	31.1	0.3	0.0	
	4	0.0	0.3	31.6	0.4	
	≥ 5	0.4	0.0	0.0	0.2	